# Chapter 9 — Subject Deep Dives

15 prompts

The rest of the book works across subjects. This chapter does not — each prompt is built around how one subject actually behaves, and what goes wrong when a general tool is pointed at it.

## WHAT'S IN HERE

| GROUP | PROMPTS | USE WHEN |
|---|---|---|
| Mathematics | 3 | Concepts, representations, error analysis |
| Science | 3 | Investigations, evidence, explanation |
| English language arts | 3 | Close reading, writing, revision |
| Social studies | 3 | Sources, corroboration, claims |
| Arts, PE and electives | 3 | The subjects general tools ignore |

## WORTH KNOWING BEFORE YOU START

**Verify everything factual in this chapter.** Historical dates, scientific values, quotations from texts, worked calculations — this is where a confident wrong answer does the most damage, because it looks exactly like a right one.

For science, never accept invented data. Predicted patterns and measured results are different things, and these prompts keep them apart.

For social studies, ask for the contested question rather than the settled summary. That is where the thinking is.

For arts and PE, be specific about equipment and space. Generic plans assume a room you do not have.

## MATHEMATICS

### SD-001 Create a concept-first mathematics explanation

Grades 3-5, 6-8, 9-12

#### WHEN YOU NEED THIS

They can do the procedure and have no idea what it means.

COPY THIS PROMPT

```
Explain [MATHEMATICAL CONCEPT] for [GRADE], starting from what it means rather
than how to do it.

Give me:
- What the concept actually is, in a sentence I could say to a [GRADE] student
- Why it exists — the problem it solves that could not be solved without it
- A concrete situation where the idea is visible before any notation appears
- Where the notation comes in, and what each part of it is standing for
- The moment students typically stop understanding and start following steps,
  and the question I ask to catch that
- Two misconceptions this explanation should prevent, and one it might create

Do not open with a definition or a rule. Do not use "just" or "simply"
anywhere.

Check every calculation you use. Flag anything you are not certain of rather
than stating it confidently.
```

BEFORE YOU SEND IT

- Work every example yourself.
- Does the notation arrive after the need for it?
- Any "just" or "simply" left?

## SD-003 Connect concrete, visual and symbolic mathematics representations

Grades K-2, 3-5, 6-8

#### WHEN YOU NEED THIS

Three representations that students never connect to each other.

#### COPY THIS PROMPT

```
Show [MATHEMATICAL IDEA] for [GRADE] across three representations.

Give me:
- Concrete: what students physically handle, using ordinary classroom
materials
- Visual: what they draw, described so I can draw it on a board
- Symbolic: the notation

Then the part that matters most:
- The explicit link between each pair — what in the drawing corresponds to
what
  in the objects, and what in the notation corresponds to what in the drawing
- The exact question I ask to make a student point at the correspondence
- Where the concrete representation breaks down, because every model has a
limit
- What a student who can do all three separately but has not connected them
  looks like

Do not present the three as three ways to get the answer. They are one idea
seen
three times.
```

#### BEFORE YOU SEND IT

- Does the concrete model use materials you actually have?
- Is the limit of the model named honestly?

## SD-004 Build a mathematics error-analysis lesson with verified solutions

Grades 3-5, 6-8, 9-12

#### WHEN YOU NEED THIS

Finding the mistake teaches more than avoiding it — if the mistakes are real.

#### COPY THIS PROMPT

```
Build an error-analysis lesson on [TOPIC] for [GRADE] mathematics.

Use fictional students — Student A, Student B — and invented work. State
clearly
that the work is fabricated for teaching.

Give me four pieces of work:
- A single conceptual error carried consistently through
- A correct method with an arithmetic slip
- A method that works here but fails in the general case
- A correct solution using an unusual but valid approach

For each: the work, what went wrong, where exactly, and what the student was
thinking. Then the answer students should give.

Then verify: recompute every piece of work and every correct answer, show your
working, and state explicitly that you have checked. Flag anything you are not
certain about.

The fourth one must be genuinely correct. An "unusual" method that is actually
wrong would teach the opposite of what I want.
```

#### BEFORE YOU SEND IT

- Recompute all four yourself, especially the "correct but unusual" one.
- Does the method-that-fails-in-general genuinely work for this case?

### SCIENCE

## SD-002 Design an evidence-based science investigation

Grades 3-5, 6-8, 9-12

#### WHEN YOU NEED THIS

Practical work where the thinking is in the design, not the worksheet.

#### COPY THIS PROMPT

```
Design an investigation into [QUESTION] for [GRADE] science.

Equipment I actually have: [LIST]
Room: [DESCRIBE — benches, sinks, space, class size]
Time: [MINUTES]

Give me:
- The question in student words, answerable with what I have
- What is varied, what is measured, and what must be held constant — with the
  ones students usually forget
- How many measurements, and why repeats matter here specifically
- The recording table, laid out exactly
- What students do with anomalous results, which is not "ignore them"
- The safety points specific to this activity, as instructions rather than
  warnings
- What we can and cannot conclude from this design

Do not invent results. Do not tell students what they should find.

Flag anything needing equipment I did not list. Do not assume a fume hood,
gas,
or running water unless I said so.

I will check all safety guidance against my school's own procedures.
```

#### BEFORE YOU SEND IT

- Check every safety point against your school's policy, not this draft.
- Does it tell students what to expect? Remove that if so.

## SD-008 Build a claim-evidence-reasoning task from supplied observations

Grades 3-5, 6-8, 9-12

### WHEN YOU NEED THIS

Students who write conclusions that are really restatements.

### COPY THIS PROMPT

```
Build a claim-evidence-reasoning task for [GRADE] science.

The observations students will have: [DESCRIBE OR PASTE THE DATA]

Give me:
- The question their claim answers
- What a strong claim looks like, and what a restatement of the data looks
like
  — students confuse these constantly
- What counts as evidence here, and which observations are relevant
- The reasoning step: the scientific idea linking evidence to claim, which is
  the part almost always missing
- A model response, and a response with good evidence and no reasoning
- Sentence frames for the reasoning step only

Use only the observations I supplied. Do not add data, and do not assume what
the results were if I have not told you.

Then tell me what a student could legitimately claim that I might mark wrong.
```

### BEFORE YOU SEND IT

- Has it added any data you did not give it?
- Consider the legitimate-but-unexpected claim before you mark.

## SD-009 Audit a science explanation for causation, uncertainty and safety

Grades 3-5, 6-8, 9-12

#### WHEN YOU NEED THIS

Explanations that are nearly right, which is the hardest kind to spot.

#### COPY THIS PROMPT

```
Audit this [GRADE] science explanation: [PASTE IT]

Tell me:
- Anywhere correlation is presented as causation
- Anywhere a hedged scientific claim has been stated as certain
- Any factual or numerical error — recompute anything computable and show your
  working
- Any simplification that will become a misconception students must later
unlearn
- Whether the level of certainty matches what the evidence supports
- Any safety-relevant statement that is incomplete or wrong
- Any anthropomorphism that will cause trouble later — atoms that "want",
  organisms that "try to"

For each, quote it and give the correction.

Where you cannot verify something independently, say so rather than agreeing
with it by default.

I will check safety statements against my school's procedures.
```

#### BEFORE YOU SEND IT

- Verify any correction it offers. It can be wrong about being right.
- The unlearn-later simplifications: decide which you accept.

### ENGLISH LANGUAGE ARTS

## SD-005 Build a close-reading lesson from an authorized supplied text

Grades 3-5, 6-8, 9-12

#### WHEN YOU NEED THIS

One text, worked properly, in its own words.

#### COPY THIS PROMPT

```
Build a close-reading lesson on the text below for [GRADE], [MINUTES] minutes.

Use only the text I paste. Do not quote from memory, do not add lines, and do
not reference a version of this text other than the one here.

Give me:
- A first read with a single narrow purpose
- A second read that goes deeper, not wider
- Four text-dependent questions from literal to inferential, each with the
exact
  lines that answer it — quoted from my text
- The two or three places students will get stuck, quoted
- The vocabulary that genuinely blocks meaning, with a gloss
- A writing or discussion task using evidence from the text

If a question cannot be answered from the text alone, say so and cut it.

TEXT:
[PASTE THE FULL TEXT — only text I am permitted to use]
```

#### BEFORE YOU SEND IT

- Check every quoted line appears in your text, exactly.
- Is the second read deeper, or just more?

## SD-006 Create an evidence-based writing task with a clear mentor model

Grades 3-5, 6-8, 9-12

WHEN YOU NEED THIS

Students who cannot picture what you are asking for.

COPY THIS PROMPT

```
Create a writing task for [GRADE] on [TOPIC OR TEXT].

Give me:
- The task, with the command word doing real work and the audience named
- A mentor model of the exact kind I am asking for, written at [GRADE] level –
  good but achievable, not a polished adult piece
- An annotation of the model: what each part does, and why it is there
- What a competent piece that misses the point looks like
- The success criteria, drawn from the model
- One sentence frame for the move students find hardest here

Rules:
- The model must be original, not a real published piece
- It should be a piece a strong student in this class could plausibly write
- Do not make it impressive at the cost of being imitable

Then tell me the risk of giving a model at all, and how to reduce it.
```

BEFORE YOU SEND IT

- Could a strong student in your class write this? If not, lower it.
- Is the model original?

### SD-007 Design a revision lesson that separates ideas, organization and conventions

Grades 3-5, 6-8, 9-12

WHEN YOU NEED THIS

Revision that is currently just checking spelling.

COPY THIS PROMPT

```
Design a revision lesson for [GRADE] writing on [TASK].

Separate three passes, in this order, and say why the order matters:
- Ideas: is there enough here, and is it the right thing
- Organisation: is it in an order a reader can follow
- Conventions: sentences, punctuation, spelling

Give me:
- What students do in each pass, and how long each gets
- One specific question per pass, not "does it make sense"
- Why conventions come last, in words I can say to a student who wants to fix
  spelling first
- What I do about the student who changes nothing
- A partner step, with what the partner is looking for — one thing only

Rules:
- Do not let conventions dominate. If the ideas are thin, spelling is not the
  problem
- Revision means changing something, not recopying it neatly
```

BEFORE YOU SEND IT

- Does the ideas pass get real time, or a token minute?
- Is the partner looking for one thing, or everything?

## SOCIAL STUDIES

## SD-010 Create a source-analysis lesson using origin, purpose and context

Grades 6-8, 9-12

WHEN YOU NEED THIS

Students who read a source as information rather than as evidence.

COPY THIS PROMPT

```
Build a source-analysis lesson for [GRADE] on this source: [PASTE IT, WITH ITS
ATTRIBUTION]

Use only the source and attribution I supplied. Do not add historical facts,
dates or context I have not given you — if context is needed, say what I
should
supply.

Give me:
- Questions on origin: who made this, when, and how we know
- Questions on purpose: who was it for, what was it trying to do
- Questions on context: what was happening that this responds to
- What this source is good evidence for, and what it is not
- The most common misreading of a source like this
- A task where students state a claim this source supports, with its limits

Do not treat the source as simply true or simply biased. Both are ways of not
reading it.
```

BEFORE YOU SEND IT

- Supply the context it asked for rather than letting it fill in.
- Check any date or fact it did use against a real reference.

## SD-011 Build a corroboration task with multiple supplied perspectives

Grades 6-8, 9-12

WHEN YOU NEED THIS

Two accounts that disagree, which is where history actually is.

COPY THIS PROMPT

```
Build a corroboration task for [GRADE] using these sources:

[PASTE EACH SOURCE WITH ITS ATTRIBUTION]

Use only what I supplied. Do not add a source, and do not supply facts from
outside these documents.

Give me:
- What the sources agree on
- Where they disagree, quoted precisely
- Questions that make students account for the disagreement rather than pick a
  winner
- What would help resolve it, and whether we have it
- A task where students write what can be said with confidence, what is
  contested, and what is unknown
- Sentence frames for writing about contested claims

Do not resolve the disagreement for students, and do not present one source as
the reliable one unless the sources themselves support that.
```

BEFORE YOU SEND IT

- Check every quotation against your source text.
- Has it quietly picked a winner?

## SD-012 Design a historical claim task that distinguishes evidence from interpretation

Grades 6-8, 9-12

#### WHEN YOU NEED THIS

The difference between what happened and what it meant.

#### COPY THIS PROMPT

```
Design a task for [GRADE] on [HISTORICAL QUESTION] using the evidence below.

EVIDENCE: [PASTE WHAT STUDENTS WILL HAVE]

Give me:
- The question, genuinely open — one where historians differ
- What the evidence establishes as fact
- What is interpretation, including interpretations I might hold without
  noticing
- Two defensible answers to the question, each with its strongest evidence
- The weakest point in each
- A task where students argue one and acknowledge the other

Rules:
- Use only the supplied evidence. Do not add facts, dates or events
- Do not present the interpretation most common in textbooks as the neutral
one
- Where the topic touches on people alive today or on contested identity, keep
  the task about the reasoning and the evidence

Then tell me which interpretation students will find easier, and why that may
not mean it is better supported.
```

#### BEFORE YOU SEND IT

- Verify every date and event it used.
- Are both interpretations genuinely defensible?

## ARTS, PE AND ELECTIVES

## SD-013 Create a skill-development lesson with modelling, practice and reflection

Grades 3-5, 6-8, 9-12

WHEN YOU NEED THIS

Skill teaching where practice is deliberate rather than just repeated.

COPY THIS PROMPT

```
Plan a [MINUTES]-minute [SUBJECT] lesson teaching [SKILL] to [GRADE].

Space and equipment I actually have: [DESCRIBE]
Class size: [NUMBER]

Give me:
- The skill broken into its components, and which one to work on today
- What I model, and the specific thing students should watch for — not "watch
  me do it"
- Practice designed so students get many attempts, not many queues
- The one cue I repeat, under five words
- What good and not-yet look like, observably, so students can self-check
- A reflection question about the process, not about enjoyment

Rules:
- No component that needs equipment I did not list
- Practice must give every student attempts, not one student demonstrating
while
  the rest watch
- Do not comment on natural ability anywhere
```

BEFORE YOU SEND IT

- Count the actual attempts each student gets. Is it enough?
- Is the cue short enough to shout across a hall?

## SD-014 Adapt a performance task for space, equipment and physical access

Grades 3-5, 6-8, 9-12

#### WHEN YOU NEED THIS

The task assumes a room, equipment or a body that not everyone has.

#### COPY THIS PROMPT

```
Adapt this [SUBJECT] task for my actual conditions: [PASTE THE TASK]

What I have: [SPACE, EQUIPMENT, TIME, CLASS SIZE]
Access needs, described as needs not students: [e.g. one student uses a
wheelchair, one has limited grip strength, one cannot be in a loud space]

Give me:
- The adapted task, achieving the same learning
- What changes, and what deliberately does not
- How a student meets the objective by a different route, with the same
standing
  — not a lesser version
- How the adaptation is unremarkable, so nobody is marked out by doing it
- What I assess, which must be identical across routes

Rules:
- Adapt the task, not the expectation
- No adaptation that removes a student from the group activity
- Do not name or describe any student
```

#### BEFORE YOU SEND IT

- Would a student feel singled out by the adapted route?
- Is what you assess genuinely the same?

## SD-015 Build an observable process rubric for an arts, PE or elective task

Grades 3-5, 6-8, 9-12

#### WHEN YOU NEED THIS

Marking the learning rather than the finished performance.

#### COPY THIS PROMPT

```
Build a rubric for [TASK] in [SUBJECT] for [GRADE], assessing [OBJECTIVE].

Weight process and technique over the finished product, and say how you have
weighted them.

Rules — these define the rubric:
- Every criterion must be observable in what the student does, not inferred
  from what they produce
- Nothing about talent, flair, natural ability, or how much they seem to enjoy
it
- Nothing about a student's body, size, or physical attributes
- Nothing requiring equipment, coaching or resources outside school
- Nothing rewarding a student who arrived already able

Give me:
- The criteria, with descriptors at proficient
- What each looks like for a student who has improved a great deal but is
still
  behind, and how that is credited
- One criterion I could observe during the lesson rather than after it

Then tell me which criterion would advantage a student with outside coaching,
and how to reduce that.
```

#### BEFORE YOU SEND IT

- Which criterion favours the already-trained student? Fix it.
- Is anything here about the body rather than the learning?